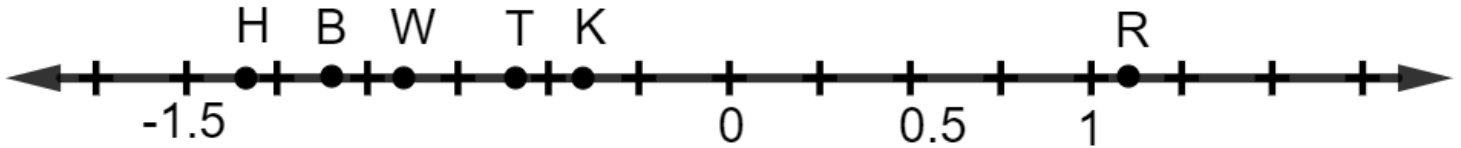


1. A number line is shown with points H, B, W, T, K, and R.



- a. Complete the statement.

The value represented by point

- ☒ B
- ☐ H
- ☐ K
- ☐ T
- ☐ W

is the opposite of the value

represented by point R because point R is the same distance

from

- ☐ -1
- ☒ 0
- ☐ 1

as point

- ☒ B.
- ☐ H.
- ☐ K.
- ☐ T.
- ☐ W.

- b. Select all the points for which the mathematical statement would be true.

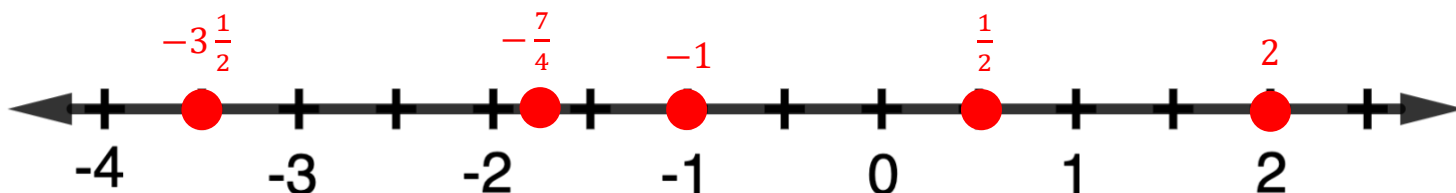
$$T > \underline{\hspace{2cm}}$$

- ☒ B
- ☒ H
- ☐ K
- ☐ R
- ☒ W

2. List the values -3.1 , $-2\frac{3}{20}$, 2.25 , $2\frac{1}{5}$, $-\frac{9}{4}$, 89% and $\frac{7}{8}$ from least to greatest.

Least to Greatest	$-3.1, -\frac{9}{4}, -2\frac{3}{20}, \frac{7}{8}, 89\%, 2\frac{1}{5}, 2.25$
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3. Plot the numbers $-3\frac{1}{2}$, 2 , $-\frac{7}{4}$, $\frac{1}{2}$, and -1 on the horizontal number line.



4. A mathematical statement that is true is shown.

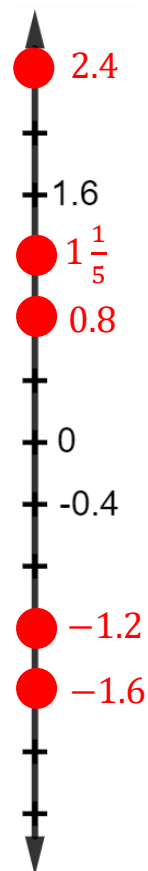
$$14\frac{7}{20} < 21\frac{3}{4}$$

Maggie used the mathematical statement to write the mathematical statement, $-14\frac{7}{20} < -21\frac{3}{4}$.

- Determine if Maggie's mathematical statement is correct by first plotting the points on a number line.
- Then, write an explanation of how the number line shows whether Maggie's mathematical statement is correct.

Number Line	
Explanation	Since both $-14\frac{7}{20}$ and $-21\frac{3}{4}$ are negative numbers, the larger value will be the value closer to 0. $-14\frac{7}{20}$ is closer to zero, so Maggie's statement is not correct.

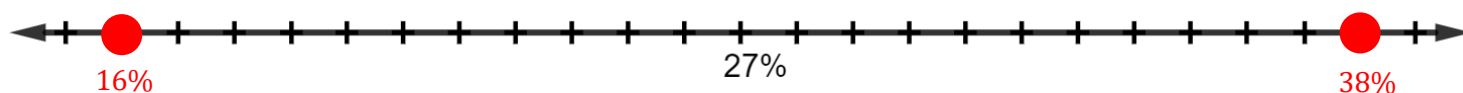
5. Plot the numbers 2.4 , -1.6, -1.2, 0.8, and $1\frac{1}{5}$ on the vertical number line.



6. In August 2005 and in June 2021, the Washington Post conducted a survey of teenagers between the ages of 14–18 years old. In 2005, 27% of teens said it was not too likely that they would be rich one day. A statement about the June 2021 survey is shown.

In 2021, the percentage of teens who said it was not too likely that they would be rich one day was 11% different from 2005.

- a. Plot the possible results on the number line.



- b. Complete the statement.

The 11% difference between the 2005 and 2021 survey

- ☐ can
☐ cannot

be

represented by absolute value because the two possible results of the 2021

survey

- ☐ are
☐ are not

the same distance from

- ☐ zero.
☐ the zero point, 27%.

7. Henry owes \$226.35 for recent purchases he made using his credit card, and he owes his parents another \$63.72.

a. Write a mathematical statement that compares the two values.

mathematical statement	$-226.35 < -63.72$
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b. Write a mathematical statement using absolute value that can be used to determine Henry's total debt.

mathematical statement	$ -226.35 + -63.72 $
-------------------------------	------------------------

c. Determine the value of Henry's total debt?

Henry's Total Debt	$\$290.07$
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8. Use the symbols $<$, $>$, or $=$ to compare each.

a. $|-65| > |-63|$

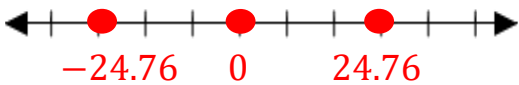
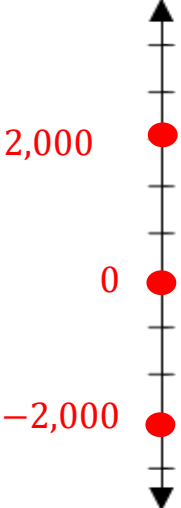
b. $83.5\% < \frac{7}{8}$

c. $-12.5 = -\frac{225}{18}$

d. $14 < |-23|$

9. For each context, complete the following in the table.

- Write the rational number that represents the situation.
- Write the opposite of the rational number.
- Represent the rational number and its opposite on the number line with appropriate labels.
- Explain the meaning of zero in the context.

	A withdrawal of \$24.76	2000 meters above sea level
Rational Number	-24.76	$2,000$
Opposite	24.76	$-2,000$
Number Line		
Meaning of Zero	Zero represents the initial amount of money in the account.	Zero represents sea level.

10. Find the value of each expression.

a. $|-42| + |-26| =$

value	68
--------------	----

b. $57 - |-26| =$

value	31
--------------	----

c. $2 \cdot |-53| =$

value	106
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